

--- Chapter 1: Origin ---

Bjarne Stroustrup created C++ in 1980 at Bell Labs.
It was made as an extension of C.
First name was 'C with Classes'.

--- Chapter 2: Growth ---

C++ added support for OOP.
Main features: classes, inheritance, polymorphism,
encapsulation and abstraction.

--- Chapter 3: Timeline ---

BCPL -> B -> C -> C with Classes -> C++

Important versions: C++98, C++03, C++11, C++14, C++17, C++20, C++23

--- Data Types ---

int: whole numbers
float: decimal numbers
double: large decimal numbers
char: single character
bool: true or false
string: text

--- Control ---

Used to control program flow.

1. if
2. if-else
3. nested if
4. switch

--- Loops ---

Used for repetition.

1. for loop
2. while loop
3. do-while loop

--- Arrays ---

Arrays store many values of same type.
Example: `int arr[4] = {5,10,15,20};`

--- Files ---

Files store data permanently.
ifstream: read from file
ofstream: write to file
fstream: read and write